

HSH HEAVY SAFE HELICOPTER

HSH-00 | REVISION 16

FINAL GENERATION 1 CONCEPTUAL / IP MASTER ENGINEERING BASELINE

Founder	Amirbahador Atai
Generation / Document	Generation 1 / HSH-00
Revision	Rev16
Maximum capsule occupancy	8 persons maximum
Vehicle target mass	26,000 kg
Safe capsule mass allocation	7,000 kg planning allocation
Main rotor target	26.0 m
Tail rotor target	4.0 m
Range target	1,000 km
Recovery study architecture	3 primary + 2 reserve canopies
Landing survival architecture	8 fixed energy-absorbing supports
Development stage	Pre-Simulation / Pre-Hardware / Pre-Prototype
Baseline status	FINAL G1 CONCEPTUAL MASTER BASELINE

Controlled status: This document is a conceptual and IP-development baseline. TARGET, ASSUMPTION, STUDY, CALC and OPEN values are not demonstrated performance. This document is not a certification approval, flight approval, production specification, or proof of achieved survival probability.

HSH HEAVY SAFE HELICOPTER

HSH-00 | REVISION 15

FINAL GENERATION 1 CONCEPTUAL / IP MASTER ENGINEERING BASELINE

Founder

Amirbahador Atai

Generation

Generation 1

Document

HSH-00

Revision

Rev15

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26.0 m

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4.0 m

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3 primary + 2 reserve canopies

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8 fixed energy-absorbing supports

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Baseline status

FINAL G1 CONCEPTUAL MASTER BASELINE

Controlled status: HSH-00 Rev15 is a conceptual and IP-development baseline. Targets, assumptions, calculations and open items are not demonstrated performance. This document is not a certification approval, flight approval, production specification, or proof of achieved survival probability.

REV15 CORRECTIVE BASELINE AND FREEZE

1. Controlled identity

The program name, document title and Generation 1 identity are unified as HSH Heavy Safe Helicopter. The controlled document is HSH-00, Revision 15, Founder: Amirbahador Atai.

2. Maximum capsule occupancy

The Generation 1 survival capsule has a hard configuration limit of 8 persons maximum. No

higher-occupancy configuration belongs to this baseline. Any future change would require formal configuration control and re-analysis of geometry, mass, CG, inertia, seating, restraints, life support, egress, structural loads, recovery sizing and certification. The 8-person limit is a configuration constraint, not a claim that eight-person crash survival has already been demonstrated. That evidence remains a future engineering objective.

3. Baseline freeze

Rev15 is the final Generation 1 conceptual master baseline. Future engineering closure is to be performed in the numbered HSH Generation 1 document tree and traced back to this baseline. A future revision of HSH-00 should occur only for a material top-level architectural or configuration change.

4. Engineering maturity

G0 Architecture Freeze is the current baseline gate. G1 through G8 remain evidence gates for analytical closure, subsystem demonstration, integrated ground demonstration, dynamics validation, structural/crash testing, recovery testing, flight development and qualification/certification.

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UNDER REV15 CONTROL

The following pages retain the Rev14 substantive engineering baseline, reflowed for readability. Rev15 corrections are limited to controlled nomenclature, the maximum-8-person capsule occupancy, and explicit baseline-status language.

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HSH GENERATION 1

HSH-00 REVISION 14

COMPREHENSIVE SAFE

HELICOPTER ENGINEERING

BASELINE

Expanded Rev13 Baseline + Engineering Maturity Framework

Founder: Amirbahador Atai

26,000 kg | Main Rotor 26.0 m | Tail Rotor 4.0 m | Capsule

7,000 kg

Status: [PROPOSED] Preliminary Controlled Engineering

Baseline

Development Stage: Pre-Simulation / Pre-Hardware /

Pre-Prototype

Revision 14 preserves the Rev13 conceptual architecture and expands the engineering, safety, verification, configuration-control and licensing framework using the Rev12 Master Engineering Document Tree as the maturity map. It does not convert targets or conceptual statements into demonstrated performance.

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HSH-00 Rev13 - Comprehensive Engineering Baseline (Full) HSH-00

REVISION 13 COMPREHENSIVE ENGINEERING BASELINE - FULL DOCUMENT

DOCUMENT CONTROL

TOP-LEVEL TECHNICAL BASELINE (REV13)

MASS ACCOUNTING - 26,000 kg (unchanged from Rev12, retained in full)

CAPSULE COMPONENT MASS - 7,000 kg (unchanged from Rev12, retained in full)

1. PURPOSE AND SCOPE

2. BASELINE STATUS AND CONFIGURATION CONTROL

3. TOP-LEVEL TECHNICAL BASELINE - NARRATIVE

4. AIRCRAFT GEOMETRY AND CONFIGURATION

5. SIX-LAYER SURVIVAL AND SAFETY ARCHITECTURE

6. DETERMINISTIC SAFETY AUTHORIZATION AND SAFETY KERNEL

7. CONTROLLED RAPID SEPARATION SYSTEM

8. OCCUPIED SURVIVAL CAPSULE

9. EIGHT FIXED SHORT MECHANICAL LANDING SUPPORTS

10. EMERGENCY STATE MACHINE

11. INDEPENDENCE, FAILURE MODES AND COMMON-CAUSE CONTROL

12. CORE SAFETY CASE - REBUILT WITHOUT MAIN-ROTOR SEPARATION (revised in Rev13)

13. STAGED SEPARATION TIMELINE (replaces Rev12's single ≤6 s target)

14. RECOVERY ARCHITECTURE - 3 PRIMARY + 2 RESERVE CANOPIES (revised in Rev13)

15. PRINCIPAL SYSTEM INTERFACES

16. VERIFICATION, VALIDATION AND EVIDENCE PROGRAM

17. ENGINEERING / IP VALUE SURFACES (narrowed and strengthened in Rev13)

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18. RISK REGISTER (new in Rev13)

19. PRIOR ART POSTURE (new in Rev13)

20. ENGINEERING BASIS AND EXTERNAL REFERENCE NOTES

21. V13 DESIGN PHILOSOPHY - CONTROLLED SURVIVAL, NOT A SINGLE DEVICE

22. DETAILED AIRCRAFT MASS BREAKDOWN (unchanged from Rev12, retained in full)

23. CRITICAL OPEN ITEMS - REV13

APPENDIX A - LICENSING PACKAGE (retained from Rev12, unchanged in substance, cross-references updated)

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HSH-00 REVISION 13 COMPREHENSIVE ENGINEERING BASELINE - FULL DOCUMENT

Founder: Amirbahador Atai

26,000 kg | Main Rotor 26.0 m | Tail Rotor 4.0 m | Capsule 7,000 kg

Revision 13 supersedes Rev12 in full. It retains every section of the Rev12 baseline and the appended licensing package, but corrects the architectural weaknesses identified in technical review - most importantly, it removes main-rotor separation from the core safety case, replaces the unsupported single-figure jettison timeline with a staged and benchmarked one, redesigns the recovery-canopy architecture, adds bottom-up rationale to the capsule mass budget, and adds two sections that were previously missing entirely: an honest Risk Register and a Prior Art Posture statement. All other narrative, mass accounting, interface definitions, requirement sets, and verification-gate structure are retained from Rev12, updated only where the corrected architecture requires it.

DOCUMENT CONTROL Field Value Document Title HSH Generation 1 -

Comprehensive Engineering Baseline Document Number HSH-00

Document Type Master Engineering Specification - Conceptual / IP Baseline

Document Class Conceptual Engineering / Intellectual Property

Development Record Revision Rev 13 - Generation 1, 26-ton-class

baseline, corrected architecture Status [PROPOSED] Preliminary Controlled

Engineering Baseline Development Stage Pre-Simulation / Pre-Hardware /

Pre-Prototype Language English Confidentiality Proprietary Concept

Development Material Founder Amirbahador Atai Engineering integrity

statement: This document defines a controlled conceptual baseline.

Numerical values marked TARGET, ASSUMPTION, STUDY, CALC or OPEN

are not demonstrated performance, certification results, or flight- proven values. Critical claims require analytical closure, simulation, ground testing, integrated testing and, where applicable, flight-test evidence. Revision intent (Rev13): Rev12 established a 26,000 kg / 26.0 m main rotor / 4.0 m tail rotor / 1,000 km range baseline with a survival-capsule and rapid-separation architecture. Rev13 preserves that numerical baseline and the survival- capsule concept in full, but corrects three architectural weaknesses that a technical reviewer or acquiring company's diligence team would flag immediately: (1) dependence of the safety case on unproven main-rotor separation, (2) an unsupported single-figure separation timeline, and (3) an under-justified 8-canopy recovery concept.

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It also adds the risk disclosure and prior-art awareness that Rev12 lacked.
TOP-LEVEL TECHNICAL BASELINE (REV13)

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Parameter Rev13 Value Status Design mass 26,000 kg [TARGET] Main rotor diameter

26.0 m

[TARGET] Main rotor disk area

530.93 m²

[CALC] Disk loading

48.97 kg/m²

[CALC] - see benchmark table §4.1 Tail rotor diameter

4.0 m

[TARGET] Tail rotor disk area

12.57 m²

[CALC] Capsule mass 7,000 kg [TARGET] - see bottom-up rationale §8

Range target 1,000 km [TARGET] Ideal hover power (study) $\approx$ 3.57 MW

[CALC] Shaft power @ FM 0.75 $\approx$ 4.76 MW [CALC] Occupancy 8

persons [TARGET / STUDY] Propulsion Two turboshaft engines [REQ /

STUDY] Main rotor separation Not part of core safety case [DEFERRED -

RESEARCH TRACK, see §12.4] Recovery canopies 3 primary + 2 reserve (5

total) [STUDY] - see §14 Command-to-clear timeline Staged per subsystem

[TARGET/OPEN per stage, see §13] MASS ACCOUNTING - 26,000 kg

(unchanged from Rev12, retained in full) Mass Group Planning Mass Status

Survival capsule, complete 7,000 kg TARGET Main airframe excluding capsule 5,000 kg ASSUMPTION / STUDY Main rotor, hub and controls 2,100 kg ASSUMPTION / STUDY Tail rotor, gearbox and drive 480 kg ASSUMPTION / STUDY Engines, transmission, gearboxes and mounts 3,100 kg ASSUMPTION / STUDY Fuel system + full usable fuel 3,400 kg ASSUMPTION / STUDY Rapid separation system 1,000 kg ASSUMPTION / STUDY Avionics, electrical, hydraulic, thermal, control 1,600 kg ASSUMPTION / STUDY Operating fluids and consumables 320 kg ASSUMPTION / STUDY Growth / configuration reserve 2,000 kg PROGRAM RESERVE TOTAL 26,000 kg CALC / CLOSURE CAPSULE COMPONENT MASS - 7,000 kg (unchanged from Rev12, retained in full)

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Capsule Component Mass Primary crashworthy cell / internal roll-cage 1,950 kg Outer capsule shell / fairing / thermal protection 500 kg Eight fixed landing supports + energy absorbers 850 kg Crashworthy seats + restraints, 8 positions 700 kg Emergency exits / doors, frames and seals 250 kg Distributed multi-canopy recovery system 650 kg Independent emergency power + batteries 250 kg Life support / oxygen / environmental 200 kg Avionics / sensing / communications / locator 250 kg Separation interfaces + release hardware 350 kg Internal secondary structure / floor / padding 500 kg Fire / smoke detection and suppression 150 kg Water landing / flotation / sealing 150 kg Inspection / access / replaceable protection 100 kg Capsule configuration growth allowance 150 kg TOTAL 7,000 kg

1. PURPOSE AND SCOPE

HSH Generation 1 defines the system-level architecture for a 26-ton-class single-main-rotor helicopter whose primary distinguishing engineering concept is an occupied survival capsule that can become the final landing boundary during a catastrophic emergency. The aircraft is designed around a layered survival chain rather than reliance on a single emergency device. Rev13 clarifies the scope in one important way relative to Rev12: the survival chain is designed to work correctly under the assumption that the main rotor is never separated from the aircraft. This is a deliberate scoping decision (see §12.4) made because no credible trajectory, gyroscopic, or structural analysis yet exists to support main- rotor separation, and building the core safety case on an unproven capability would undermine confidence in the entire architecture. The baseline covers the aircraft, occupied survival capsule, rapid selective separation

system (non-rotor components), safety initiation paths, deterministic safety authorization, stabilization, recovery systems, impact attenuation, post-event survival and the interfaces required to verify the complete chain. The design philosophy is not “separate everything whenever the aircraft starts descending.” Separation is a controlled emergency mode authorized only when defined conditions are satisfied. The survival architecture is intended to support multiple emergency outcomes: conventional flight recovery where possible, autorotation where applicable, selective hazard/mass reduction, capsule separation, parachute-assisted descent, energy-absorbing landing and post-landing rescue. The architecture must not require the most extreme survival mode

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for every failure. The baseline also preserves human emergency authority. Automatic systems may initiate or recommend a survival sequence only within defined authorization envelopes; the crew must retain an independent means to command, inhibit or terminate automatic action where technically and physically possible. This document is an engineering/IP baseline, not a certification approval. Airworthiness, structural substantiation, parachute qualification, release reliability, occupant injury criteria, flight performance and global patent novelty remain open work items.

2. BASELINE STATUS AND CONFIGURATION CONTROL

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Status Label Meaning in this document [LOCKED CONCEPT] Architecture intentionally retained as a baseline design principle; implementation remains subject to engineering verification [REQ] Top-level requirement proposed for subsequent requirement allocation [TARGET] Numerical or functional development objective; not demonstrated [ASSUMPTION / STUDY] Working engineering assumption used for preliminary sizing/accounting [CALC] Value calculated from stated assumptions and equations [OPEN] Requires analysis, simulation, test, supplier data or design decision [OPEN - CRITICAL] An open item with material effect on the core safety case; flagged for priority closure (new in Rev13) [DEFERRED - RESEARCH TRACK] Explicitly excluded from the licensable core baseline pending independent research (new in Rev13) [TBD - DEPENDENCY] Cannot be responsibly fixed until a dependent design activity closes

[PRIOR ART STATUS: NOT YET SEARCHED] No novelty conclusion is implied. The baseline shall be configuration-controlled. Any change to MTOW, rotor diameter, capsule mass, occupant count, separation sequence, parachute architecture, landing-support geometry or safety-layer independence shall trigger an impact assessment across mass properties, CG, inertia, dynamics, structures, controls, recovery and verification.

3. TOP-LEVEL TECHNICAL BASELINE - NARRATIVE

Maximum Take-Off / Design Mass is 26,000 kg [TARGET]. Occupancy is 8 persons [TARGET/STUDY]. Flight architecture is a conventional single main rotor plus tail rotor [REQ/STUDY]. Main rotor diameter is 26.0 m [TARGET] with disk area 530.93 m² [CALC] and weight-based disk loading 48.97 kg/m² [CALC] - see the benchmark comparison in §4.1, which situates this figure against real heavy-lift rotorcraft rather than leaving it unexplained. Tail rotor diameter is 4.0 m [TARGET] with disk area 12.57 m² [CALC]. Propulsion architecture is two turboshaft-class engines [REQ/STUDY]. Landing gear is not retractable wheeled gear; instead, eight short, thick fixed mechanical supports integrated with the capsule [TARGET/CONCEPT]. Range target is 1,000 km [TARGET]. Hover endurance is to be derived after power/fuel/thermal closure [OPEN]. The survival capsule is the dedicated occupied final landing boundary [REQ], with a preliminary complete mass of 7,000 kg [TARGET] and bottom-up rationale added in §8. Capsule recovery in Rev13 uses a 3-primary/2-reserve distributed canopy architecture [STUDY] rather than Rev12's undifferentiated 8-canopy concept - see §14 for the rationale. Rapid

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separation covers selective separation of designated high-mass/fire-hazard components excluding the main rotor [REQ/CONCEPT]. The separation development objective is expressed as a staged, per-subsystem timeline rather than a single figure [TARGET/OPEN per stage - see §13]. The 26.0 m rotor is intentionally sized for the 26-ton class; the resulting disk loading of approximately 48.97 kg/m² is, per §4.1, a deliberate hover-margin and downwash-severity trade rather than an unexamined byproduct.

Preliminary ideal hover power is approximately 3.57 MW at sea-level standard density; applying an illustrative figure-of-merit assumption of

0.75 gives approximately 4.76 MW rotor power - a study calculation only.

Installed engine power must be determined from a complete power-required/power-available model including profile power, induced power, transmission losses, control margins, temperature, density altitude, rotor speed, transients and OEI requirements. The 1,000 km range is a

design target; it shall not be presented as achieved range until the mission model closes with aircraft drag, rotor performance, engine specific fuel consumption, fuel capacity, reserves, payload, temperature/altitude and operational constraints.

4. AIRCRAFT GEOMETRY AND CONFIGURATION

The geometry below is a preliminary architecture sized around the 26-ton class. Dimensions other than rotor diameters are deliberately kept as ranges or open items until aerodynamic, structural, cabin, capsule and systems layouts are reconciled.

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Geometry Item Preliminary Definition Status Overall length Approx.

16.5-18.0 m [TARGET/STUDY] Maximum fuselage width

Approx. 4.0-4.5 m

[TARGET/STUDY] Overall height Approx. 4.8-5.3 m [TARGET/STUDY] Main rotor diameter

26.0 m

[TARGET] Tail rotor diameter

4.0 m

[TARGET] Rotor arrangement Single main rotor + tail rotor [REQ/STUDY]

Landing supports 8 fixed short mechanical supports under capsule

[TARGET] Retractable landing gear None [LOCKED CONCEPT] Capsule

position Central occupied boundary integrated within primary airframe

[LOCKED CONCEPT] Capsule normal-flight load path Dedicated primary

attachment system with controlled emergency release [REQ] Capsule

emergency load path Capsule becomes independent final landing element

after verified clearance [REQ] The eight landing supports are not simple

struts attached to a thin capsule shell. The intended load path is: ground

contact → foot/ground interface → progressive energy absorber → lower

capsule load path → primary capsule structure → occupant restraint/seat

energy management. Structural sizing shall consider vertical, longitudinal,

lateral and torsional loads, slope, uneven terrain, soft soil, asymmetric

touchdown and failed-leg cases.

4.1 Rotor Disk Loading - Benchmark Comparison (new in

Rev13)

Rev12 presented the 48.97 kg/m² disk loading figure without context,

which invites a reviewer to interpret it as either an error or an unexplained

anomaly. Rev13 adds the comparison a diligence engineer will make

regardless, and states the design rationale explicitly: Aircraft MTOW (approx.) Main Rotor Diameter Disk Loading (approx.) Mi-26 56,000 kg

32.0 m

~55 kg/m² CH-53K 38,400 kg

24.1 m

~58-60 kg/m² HSH Gen1 (Rev13) 26,000 kg

26.0 m

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48.97 kg/m²

Design rationale: HSH's disk loading is deliberately lower than comparable heavy rotorcraft. This is presented as a trade for greater hover power margin - useful given the added mass and complexity of the capsule and separation systems - and for reduced downwash severity near the fixed landing supports during ground operations. This framing turns what would otherwise be a flagged anomaly into a stated, defensible engineering decision, to be confirmed or revised once Document 063 (Main Rotor Disk Loading Performance Study) closes.

5. SIX-LAYER SURVIVAL AND SAFETY ARCHITECTURE

The architecture retains the six-layer functional survival philosophy. The layers are functionally distinct. Duplication alone shall never be used as evidence of independence. Independence shall be assessed for power, signal routing, sensors, processors, physical location, structural vulnerability, fire, EMI/lightning, common-cause failures and human interaction.

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Layer Function Primary Purpose L1 Primary automatic emergency detection and initiation Detect catastrophic states and initiate a validated survival sequence when authorization conditions are met L2 Independent backup initiation Provide a second physically/electrically segregated automatic path if L1 is unavailable L3 Independent manual/crew authority Allow crew initiation, inhibit and emergency authority without dependence on L1/L2 L4 Rapid selective physical separation Separate designated non-rotor mass/fire hazards from the capsule using controlled release mechanisms L5 Direct-impact survival Protect occupants through capsule structure, restraints, seats and eight energy- absorbing supports L6

Recovery and post-event survival Deploy recovery system, stabilize landing, locate occupants and support post-landing survival/rescue The design principle is: AI or advanced prediction may assist detection, classification and decision support; deterministic safety logic authorizes safety-critical actuation. Confidence or model output shall never be treated as equivalent to safety authorization. This principle underlies the strongest IP claim in the package - see §17, IP-A.

6. DETERMINISTIC SAFETY AUTHORIZATION AND SAFETY KERNEL

The Safety Kernel is the deterministic authorization boundary between emergency detection/decision logic and physical survival actuation. Its purpose is to prevent a computationally complex or uncertain decision layer from directly commanding hazardous separation without satisfying explicit safety preconditions. The Safety Kernel shall evaluate, as applicable: altitude above terrain and minimum recovery altitude; airspeed and dynamic pressure envelope; pitch, roll, yaw and angular-rate limits; rotor RPM, rotor state and rotor control state; engine/fire/fuel hazard state; structural health and confirmed structural failure state; capsule attachment health and release readiness; separation trajectory clearance; potential collision between released components and capsule/parachutes; parachute deployment envelope and recovery-system availability; landing support availability and degraded configurations; sensor validity and disagreement; power availability for release and recovery systems; communication availability where applicable; manual crew authority and emergency override state; rollback/fallback availability. If the safety conditions are not satisfied, the system shall inhibit the unsafe action or transition to a defined degraded mode. Safety and physical feasibility take

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priority over optimization or mission continuation. No software model, AI classifier, optimizer or predictive system shall bypass the Safety Kernel, aircraft protection systems, interlocks, equipment limits or human emergency authority.

7. CONTROLLED RAPID SEPARATION SYSTEM

7.1 Separation philosophy

The rapid separation system is a selective, staged, controlled mechanism for reducing the mass, fire load and rotating/energetic hardware that must remain associated with the occupied capsule during the final descent. It is not a blanket requirement to release every aircraft component in every emergency. The system shall use state-dependent release logic. Components shall be designated as jettisonable only after structural,

aerodynamic, dynamic, fire, control, trajectory and collision analysis demonstrates that release is acceptable within a defined envelope.

7.2 Candidate separable components (revised in Rev13)

Main engines and associated high-mass engine installation interfaces, where practical. Selected transmission/drive components where release is structurally and dynamically acceptable. Tail rotor, tail-rotor gearbox and drive shafting where release is safe. Primary fuel tanks and major fuel lines that constitute a significant fire hazard, after safe fuel isolation. Selected external structural panels or high-mass attachments that do not form part of the capsule primary load path. Other components only through configuration-controlled hazard and trajectory analysis. Main rotor blades and hub components are explicitly excluded from the core candidate list in Rev13. See §12.4 for the rationale and for the deferred research-track status of this capability.

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7.3 Mandatory release protections

Positive mechanical retention during normal flight; electrical/logic inhibits against inadvertent release; independent command paths; defined arm/fire sequence; release-state sensing; fuel and energy isolation before or coordinated with release; asymmetric/partial-release detection; post-release trajectory monitoring where practical; capsule-clear verification before parachute authorization; defined failed-release and rollback/fallback behavior.

8. OCCUPIED SURVIVAL CAPSULE

The occupied survival capsule is the central safety boundary of HSR Generation 1. During normal flight it is structurally integrated into the aircraft through controlled primary attachments. During a validated catastrophic survival sequence, it is intended to become an independent occupied landing element. Capsule Parameter Definition Status Occupancy 8 persons [TARGET/STUDY] Complete capsule mass 7,000 kg planning target [TARGET] Primary structure Continuous CFRP roll-cage with metallic/titanium nodes and controlled joints [CONCEPT] Seats Energy-absorbing seats with five- point/appropriate restraints [TARGET] Landing supports 8 short, thick fixed mechanical supports with progressive energy absorption [TARGET] Parachute architecture 3 primary + 2 reserve distributed canopies [STUDY - revised, see §14] Capsule emergency systems Independent emergency power, sensing, locator, communications and essential life- support [TARGET] Capsule clearance Verified before

recovery deployment [REQ] Post-landing stability Eight-point support architecture with asymmetric/failed-leg analysis [REQ] The capsule shall be designed as a real structural system rather than a passenger cabin merely surrounded by a shell. The engineering program shall address load paths, crash loads, fatigue, damage tolerance, joints, attachment fittings, seat loads, floor loads, fire/smoke, environmental exposure, water/ditching provisions where required, emergency egress and occupant injury criteria. The capsule shall retain sufficient independent capability after separation to support recovery and rescue. Essential systems should not depend on the main airframe power bus after separation.

8.1 Capsule mass - bottom-up rationale note (new in Rev13)

The 7,000 kg figure is retained from Rev12 as a planning target pending the bottom-up closure defined in Document 113 of the Master Document

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Tree, but Rev13 adds the context a buyer or reviewer needs to judge its plausibility rather than leaving the number unexplained: Per-occupant mass at 8 occupants is approximately 580-700 kg - materially higher than a standard aviation cabin allocation, because the capsule mass includes a fully independent structural cell (not shared load path with the airframe), the eight-point external landing/energy-absorption interface, distributed-parachute hard points, and independent emergency power and life support, none of which a conventional cabin carries. The nearest relevant structural analog is a crewed spacecraft return capsule, where fully-equipped per-occupant mass (including independent life support and thermal protection) commonly runs in a comparable or higher range - a closer comparison class than a helicopter cabin. This remains an explicit planning allocation, not a verified figure; Document 113 (Capsule 7000 kg Mass Budget Bottom-Up) is still required before this number can be presented as engineering-closed.

9. EIGHT FIXED SHORT MECHANICAL LANDING SUPPORTS

HSH Generation 1 uses eight short, thick, fixed mechanical supports located below the survival capsule. The concept is intentionally different from conventional retractable landing gear: the supports are part of the survival capsule's impact-attenuation system and remain with the occupants after separation. Eight supports provide load distribution and redundancy against uneven terrain and partial failure. Each support shall include a defined mechanical energy-absorption function rather than relying solely on elastic deformation. Replaceable or inspectable sacrificial/crush elements should be considered to improve post-event maintainability. The attachment system shall transmit controlled loads into

the capsule's primary lower structure. The system shall be analyzed for vertical, lateral, longitudinal and combined impact loads. Asymmetric landing with one or more failed/degraded supports

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is a required analysis case. Terrain cases shall include hard flat ground, sloped ground, uneven ground and soft/low- bearing-strength surfaces. Ground clearance and rotor-downwash interaction shall be evaluated. Post-landing stability, tipping resistance and rescue access shall be included. Preliminary capsule impact energy follows $E = 1/2 m v^2$. For a 7,000 kg capsule, illustrative kinetic energies are approximately 126.0 kJ at 6 m/s, 224.0 kJ at 8 m/s and 350.0 kJ at 10 m/s. These are not allowable impact energies; they are sizing illustrations only. Actual survival depends on energy absorption, occupant pulse, attitude/rates, terrain and structural response.

10. EMERGENCY STATE MACHINE

HSH Generation 1 shall use an explicit emergency state machine rather than a single binary "normal/emergency" flag:

1.

NORMAL FLIGHT - all safety layers monitored; release hardware mechanically locked.

2.

ANOMALY DETECTED - sensor/model detects a potential abnormal condition.

3.

CONFIRMATION - independent evidence and validity checks determine whether the condition is real.

4.

IMPACT ASSESSMENT - estimate control authority, altitude, airspeed, fire/structural risk, rotor/engine state and available recovery modes.

5.

RECOVERY ATTEMPT - use conventional control, engine redundancy or autorotation where safe and appropriate.

6.

SURVIVAL AUTHORIZATION - Safety Kernel evaluates whether selective separation/capsule recovery is permitted.

7.

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STAGED SEPARATION - execute only the designated release set for the specific emergency state (non-rotor components only, per §7.2).

8.

CAPSULE CLEARANCE - verify physical separation and identify any attached components.

9.

RECOVERY DEPLOYMENT - authorize parachute/recovery system within validated envelope.

10.

DESCENT MANAGEMENT - monitor attitude, rates, canopy state and landing-support readiness.

11.

LANDING ATTENUATION - deploy/condition energy absorbers as required.

12.

POST-LANDING SURVIVAL - stabilize, locate, communicate, protect occupants and support rescue.

13.

AUDIT - preserve event data, release states, sensor states and command history for engineering analysis. Representative state-dependent behavior: an isolated engine failure should not automatically trigger capsule separation; a major fire may prioritize fuel isolation and selective separation; loss of control may require a different separation envelope; structural damage may require immediate transition; a combined catastrophic event may require the full survival chain.

11. INDEPENDENCE, FAILURE MODES AND COMMON-CAUSE

CONTROL

The survival architecture introduces new hazards as well as new protections. The following failure classes shall be treated as explicit design cases:

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Failure / Hazard Required Design Response Unintended separation
Mechanical locks, electrical inhibits, command validation, monitoring and
quantitative safety objective Partial separation State sensing,

asymmetric-load analysis and defined degraded recovery mode
Asymmetric separation Dynamic analysis, CG/inertia update and
post-release control assessment Release mechanism failure Independent
path or defined safe fallback; no assumption of perfect release Sensor loss
/ bad data Validity monitoring, redundant/diverse sensing and safe-state
logic Automatic detection failure Independent backup and manual crew
initiation Power-bus failure Independent survival power where required for
critical functions Communication loss Local deterministic safety logic;
communication shall not be the sole authorization path AI/model failure
Deterministic safety logic remains authoritative Fire during separation Fuel
isolation, fire containment and release sequencing Released component
collision Trajectory analysis, clearance envelopes and controlled
sequencing Parachute deployment outside envelope Hard inhibits and
validated deployment conditions Landing-support failure Eight-support
redundancy and asymmetric/failed-leg structural cases Common-cause
EMI/lightning/fire/structural damage Dedicated common-cause analysis
and physical/electrical segregation Independence is not established merely
because two boxes, sensors or switches exist. The verification program
shall assess common power, common wiring routes, common structure,
common software, common sensor physics, common fire zones and
common environmental vulnerabilities.

12. CORE SAFETY CASE - REBUILT WITHOUT MAIN-ROTOR SEPARATION

(revised in Rev13)

12.1 Why this changed

Rev12 listed the main rotor as a “candidate separable component,”
conditional on future analysis. In practice, this created an implicit
dependency: several narrative passages in Rev12 implied that in the most
catastrophic emergency states, full survival relied on the possibility of
rotor separation. A 26 m rotor carries enormous rotational kinetic energy
and gyroscopic moment; no credible 6-DOF or structural trajectory analysis
yet exists to show that severing and clearing it from a descending capsule
and its parachutes can be done safely. Leaving this dependency in the

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baseline gives any competent technical reviewer grounds to discount the
rest of the document’s rigor.

12.2 The corrected principle

The survival chain is designed to work assuming the main rotor is never
separated.

12.3 Revised primary survival chain

1.

Controlled flight recovery / autorotation (unchanged from Rev12).

2.

Selective separation of non-rotor hazards only: engines, fuel tankage (post-isolation), tail rotor/drive assembly, designated fire-load panels.

3.

Capsule release from the airframe - the capsule departs from beneath/behind the still-attached (or autorotating) main rotor disk, not through it.

4.

Capsule-only descent under the distributed recovery canopy system (§14).

5.

Impact attenuation via the eight fixed supports (§9).

6.

Post-landing survival and locate/rescue (§16).

12.4 Main-rotor separation - deferred research track

Main-rotor separation is retained in the document as a labeled, walled-off future research item, explicitly excluded from the licensable core baseline and from the primary IP claims (§17). This is a deliberate configuration-control decision: it prevents a buyer's diligence team from being able to discount the entire package on the strength of its single weakest claim, while preserving the option to reintroduce a rotor-separation capability later if and when independent 6-DOF, structural and trajectory analysis (Document 080, HSH Rotor Emergency Release Feasibility Study) demonstrates it is

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achievable.

13. STAGED SEPARATION TIMELINE (replaces Rev12's single ≤6 s target)

Rev12 proposed a single "≤6 s from validated command to verified capsule-clear state" target with no supporting rationale. Rev13 replaces this with a staged, per-subsystem timeline that shows its work and flags its own weakest link rather than hiding it: Stage Action Target Window Confidence Basis S0 Validate emergency state; arm authorized release path; isolate fuel/electrical/energy hazards

0.0-0.5 s

Comparable to modern ejection- seat arming logic S1 Fuel isolation and designated fire-hazard separation

0.5-1.8 s

Comparable to certified crash- fuel-shutoff systems in existing rotorcraft S2 Engine and tail-rotor drive separation (non-main-rotor)

1.8-3.5 s

Comparable to external cargo- release hardware timing, scaled up S3 Capsule-to-airframe release

3.5-5.0 s

[OPEN - CRITICAL] - no direct engineering analog exists; flagged honestly rather than asserted S4 Capsule clearance verification

5.0-5.8 s

Sensor-response-limited, not mechanism-limited S5 Recovery canopy authorization After S4, gated on verified clearance Not time-gated; correctness prioritized over speed If detailed architecture cannot safely achieve these windows, targets shall be revised under configuration control rather than achieved by reducing verification rigor. Partial and asymmetric separation remain first-class emergency states at every stage - the system shall not assume that every release mechanism functions perfectly.

14. RECOVERY ARCHITECTURE - 3 PRIMARY + 2 RESERVE

CANOPIES (revised)

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in Rev13) Rev12 proposed an undifferentiated 8-canopy distributed system with no stated failure-tolerance rationale. Rev13 replaces this with a structure closer to flight-proven capsule-recovery precedent: Recovery Mode Use Case Status Controlled flight recovery Recoverable control/propulsion disturbance [REQ/STUDY] Autorotation Engine/power loss when altitude/rotor state permit [REQ/STUDY] Selective mass/hazard separation Fire, structural or combined emergency requiring reduction of carried hazard/mass [REQ/CONCEPT] Capsule separation Catastrophic state where independent occupied recovery is preferable [REQ/CONCEPT] 3 primary recovery canopies Primary post-separation vertical recovery [STUDY - revised from 8 to 3+2] 2 reserve canopies Deployed only on detected primary-canopy failure, not by default [STUDY - new in Rev13] Eight energy-absorbing supports Final ground impact attenuation [REQ/TARGET] Optional airframe recovery parachute Residual-airframe recovery or unmanned asset protection [OPTION/TRADE STUDY] Post-landing survival Locator, communications, lighting, emergency power

and rescue interface [REQ] Rationale for the change: three larger primary canopies, sized to the 7,000 kg capsule using standard aerospace descent-rate targets, are directly preceded by crewed spacecraft return-capsule practice (three-main architectures at comparable or greater mass). Fewer, larger canopies reduce inflation-sequencing complexity and inter-canopy collision risk at low altitude - the failure mode most likely in a real emergency, since most rotorcraft accidents occur during low- altitude phases of flight. Two independent reserve canopies provide redundancy without multiplying simultaneous- inflation risk. This directly addresses the weakest point of Rev12's recovery concept. Explicit open item retained: the minimum deployment altitude at which this (or any) capsule recovery system can be

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relied upon is [OPEN - CRITICAL] . Rev13 does not claim this is solved; it states plainly that low-altitude emergencies may fall outside any capsule-recovery envelope regardless of canopy count, and that this must be closed analytically (Document 159, Capsule Parachute Sizing Analysis) before any survivability claim is made for low-altitude scenarios.

15. PRINCIPAL SYSTEM INTERFACES

Interface Definition Control Requirement CAP-AF Capsule ↔ airframe structural attachment Normal-flight load path + controlled emergency release + positive retention CAP-PAR Capsule ↔ parachute system Deployment only after validated capsule state and clearance CAP-LG Capsule ↔ eight landing supports Primary crash/landing load path with energy absorption JET-PWR Independent release power Segregated from common single-point power failure where required JET-CMD Automatic/backup/manual commands Independent paths and protected actuation JET-SNS Release-state sensing Positive confirmation of attached/released state FUEL-ISO Fuel isolation ↔ separation Isolation before or coordinated with release according to hazard analysis REC-LOC Post-landing locator/communications Independent emergency power and antenna strategy AV-CTRL Avionics/control ↔ Safety Kernel Deterministic authorization boundary All interfaces crossing survival boundaries shall be documented in Interface Control Documents and assessed for common-cause vulnerability. Connector, harness, hydraulic, fuel and structural interfaces shall be designed so that emergency separation does not create an uncontrolled hazard.

16. VERIFICATION, VALIDATION AND EVIDENCE PROGRAM

The following evidence gates are required before any high-confidence safety or performance claim is made: Gate Evidence Objective G0 Architecture Freeze - confirm safety philosophy, system boundaries, interfaces and emergency states G1 Analytical Closure - mass, CG, inertia, rotor sizing, power, fuel, structural load paths and separation kinematics G2 Subsystem Demonstration - release mechanisms, locks/inhibits, energy absorbers, parachute components, sensors and independent power G3 Integrated Ground Demonstration - end-to-end sequence without flight exposure; verify timing, state sensing and capsule clearance G4 Dynamic / 6-DOF Simulation - vehicle and capsule dynamics through representative emergency states, partial/asymmetric release and parachute inflation G5

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Structural / Crash Testing - capsule, attachment points, seats, landing supports, energy absorbers and occupant-protection evidence G6 Recovery System Testing - canopy deployment, inflation loads, failure cases, stability and landing performance G7 Flight Development - controlled flight tests within approved envelope; validate normal and emergency recovery behavior G8 Certification / Qualification - evidence package against the selected certification basis and applicable standards Terms shall be used carefully: “specified” means defined in the design; “analyzed” means a documented analytical result; “simulated” means supported by a model; “tested” means demonstrated in a defined test; “verified” means evidence shows compliance with a requirement; “validated” means suitability has been established for the intended use; “qualified” and “certified” require the applicable formal processes. Rev13 note: as of this revision, no gate beyond G0 (Architecture Freeze) is complete. This is stated plainly rather than

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implied otherwise - see §18, Risk Register.

17. ENGINEERING / IP VALUE SURFACES (narrowed and strengthened in

Rev13) Rev12 listed nine candidate IP elements at roughly equal weight, which dilutes attention and invites a diligence team to discount the whole set once the weakest are found unpatentable or over-broad. Rev13 concentrates on the three most defensible claims and retains the rest as supporting system narrative rather than standalone claims.

17.1 Primary claim candidates

IP-A - Deterministic Safety Kernel gating AI-assisted emergency prediction from physical actuation. The combination of (a) AI/ML-based anomaly prediction, (b) a hard deterministic precondition layer, and (c) selective, state-dependent (not all-or-nothing) actuation is a specific systems-engineering combination, not a single known device. This is assessed as the strongest claim in the package. IP-B - Eight-point fixed capsule-integrated landing support with independent asymmetric/partial-failure state management. Combining the structural configuration (eight-point, capsule-integrated, non-retractable, energy-absorbing) with explicit failure-state detection and degraded-recovery logic as a single claim is narrower and more defensible than treating the hardware and the logic as separate elements. IP-C - State-dependent, non-rotor selective separation sequencing for an occupied survival capsule. By explicitly excluding main-rotor separation (§12), this claim is narrowed to something closer to precedented “selective jettison” systems with a specific new element: capsule-centric, staged, state-dependent sequencing logic (§13) rather than an unbounded claim over “separate anything.”

17.2 Supporting system narrative (not standalone patent candidates)

The six-layer survival taxonomy (§5), the multi-mode recovery chain (§14), state-aware partial/asymmetric-release management, the optional residual-airframe recovery concept, and the immutable emergency event record are retained as descriptive system architecture. They strengthen the buyer’s understanding of the overall product but are not presented as independently patentable - a narrower, honestly-scoped claim set is more credible to patent counsel than nine equally-weighted assertions.

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18. RISK REGISTER (new in Rev13)

Risk Severity	Current Status	Capsule-to-airframe release (Stage S3, §13)
	has no direct engineering precedent	High Open research item, explicitly flagged
	Low-altitude canopy deployment envelope may exclude many real	
	accident profiles	High Open, needs minimum-altitude closure study
	(Document 159)	7,000 kg capsule mass not yet bottom-up verified
	Medium Planning allocation only, Document 113 pending	No 6-DOF simulation
	evidence exists yet for any separation stage	High Gate G4 not started
	No professional prior-art search has been conducted	Medium See §19
	Certification basis (CS-29 analog) not formally engaged with any regulator	
	Medium Conceptual reference only	Main-rotor separation capability, if later

reintroduced, remains structurally unproven Medium Deferred to research track, §12.4 A buyer's engineering diligence team will identify these risks regardless of whether they are disclosed. Naming them here, in the founder's own words, is what makes the surrounding technical claims credible rather than promotional.

19. PRIOR ART POSTURE (new in Rev13)

This section does not assert novelty conclusions - that requires a professional prior-art search - but names the nearest known analog classes so a reviewer sees baseline awareness rather than legal naivety:

Ejection-seat-equipped rotorcraft. Existing rotorcraft designs (e.g., certain Soviet/Russian attack helicopters) use rotor-blade-severing explosive charges paired with a crew ejection seat. This is directly relevant prior art for

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any main-rotor-related separation claim, which is precisely why Rev13 removes main-rotor separation from the core claim set (§12, §17). Crewed spacecraft return capsules. Multi-canopy recovery, crashworthy seating and independent post-landing life support in existing crewed spacecraft are relevant prior art for the capsule-recovery and life-support elements of this baseline. Rev13's IP language (§17) is narrowed to the rotorcraft-specific combination rather than the general capsule-recovery concept, which is not novel on its own. Existing crash-attenuating landing gear / energy-absorbing strut systems in certified rotorcraft are relevant to the eight-support claim (IP-B). Novelty must rest on the capsule-integration and failure-state-management aspects specifically, not on energy-absorbing landing gear in general. A professional prior-art search against IP-A, IP-B and IP-C specifically (not the whole document) remains a prerequisite before any patent filing or broad data-room disclosure.

20. ENGINEERING BASIS AND EXTERNAL REFERENCE NOTES

The baseline is derived primarily from the founder's prior LSH Document 00 Rev3.2 and the higher-maturity rapid- jettison Generation 1 document. The previous baseline established the six-layer survival philosophy, independent automatic/manual initiation, controlled separation, eight landing supports, the capsule-as-final-landing-boundary concept, and the need to treat partial/asymmetric separation as first-class states. Rev12 updated the Generation 1 planning point to 26,000 kg, 26.0 m main rotor,

4.0 m tail rotor and 1,000 km range target. Rev13 preserves
this numerical

baseline in full and corrects the architectural dependencies described above. Current EASA CS-29 material was reviewed as an external engineering reference. The EASA material emphasizes that jettisonable configurations require satisfactory separation characteristics, including clear trajectory, no damaging interaction, no adverse/uncontrollable aircraft reaction, acceptable post-jettison stability/control and acceptable post-jettison performance; the verification program therefore treats these as core analysis/test concerns rather than assuming safe release. External reference: EASA, Easy Access Rules for Large Rotorcraft (CS-29), Amendment 12, March 2026.

21. V13 DESIGN PHILOSOPHY - CONTROLLED SURVIVAL, NOT

A SINGLE

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DEVICE The defining engineering proposition of HSH Generation 1 remains the integration of multiple survival mechanisms into one state-aware architecture. The aircraft is not intended to depend solely on a parachute, solely on autorotation, solely on a detachable cabin or solely on automatic software. Rev13 strengthens this proposition by removing the one dependency (main-rotor separation) that most directly contradicted it: a safety case that quietly needs its riskiest, least-proven capability to succeed is not actually a layered safety case. The intended chain is: PREVENT/MONITOR → DETECT → CONFIRM → ASSESS → RECOVER IF POSSIBLE → SAFETY AUTHORIZE → SELECTIVELY SEPARATE (NON-ROTOR) → VERIFY CAPSULE CLEARANCE → DEPLOY RECOVERY → MANAGE DESCENT → ABSORB IMPACT → SURVIVE/LOCATE/RESCUE → AUDIT. The strongest future evidence will come from a closed mass model, rotor/propulsion analysis, 6-DOF simulation, deterministic safety logic, separation mechanism tests, capsule structural tests, parachute tests and progressively integrated demonstrations. Document disposition: Rev13 is suitable as the controlled conceptual/master engineering baseline for the next engineering phase, and is materially more credible to a technical acquirer than Rev12 because it discloses its own weakest points rather than presenting an appearance of completeness. It should not yet be presented as a certified aircraft specification or as proof of achieved 26-ton performance, 1,000 km range or survival probability.

22. DETAILED AIRCRAFT MASS BREAKDOWN (unchanged from

Rev12, retained

in full) Main Airframe - 5,000 kg: Fuselage primary load-bearing structure 1,450 kg; frames, bulkheads and longerons 650 kg; tail boom and tail structural assemblies 520 kg; engine and transmission support structure

380 kg; rotor mast/transmission support integration structure 260 kg; external skins, fairings and aerodynamic panels 310 kg; doors, access panels and hinges 190 kg; firewalls, barriers and structural fire protection 170 kg; structural fittings, lugs, fasteners and local reinforcements 260 kg; cabin/service floor structure outside capsule boundary 210 kg. Main Rotor System - 2,100 kg: Main rotor blades (26.0 m system) 980 kg; rotor hub and blade retention/grips 280 kg; mast and rotor-head structural components 260 kg; swashplate and pitch-control hardware 170 kg; pitch links, control rods and bearings 110 kg; rotor vibration/dynamic-control hardware 80 kg; rotor lubrication/inspection/access hardware 40 kg; configuration allowance 40 kg. Tail Rotor System - 480 kg: Tail rotor blades (4.0 m system) 145 kg; tail rotor hub and blade retention 65 kg; tail rotor gearbox 95 kg; drive shafting, couplings and bearings 60 kg; tail rotor

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controls and pitch mechanism 35 kg; tail structure/interface hardware 25 kg.

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Engines and Main Transmission System - 3,100 kg: Engine #1 complete installed 700 kg; Engine #2 complete installed 700 kg; engine accessories and engine-control hardware 180 kg; main gearbox/combining transmission 650 kg; intermediate/reduction transmission elements 180 kg; drive shafts and couplings 100 kg; lubrication, oil cooling and filtration 80 kg; engine/gearbox mounts and vibration isolation 60 kg; exhaust and hot-section installation hardware 50 kg. Fuel System + Full Usable Fuel - 3,400 kg: Usable fuel mass 2,100 kg; fuel tanks and crash/fire-resistant tank structure 280 kg; fuel pumps and boost/transfer units 90 kg; fuel lines, hoses, valves and fittings 110 kg; fuel filters, sensors and quantity measurement 55 kg; fuel isolation/emergency shutoff hardware 75 kg; venting, surge and inerting/fire-protection provisions 70 kg; unusable/residual fuel allocation 120 kg. Rapid Selective Separation System - 1,000 kg: Primary capsule release mechanisms and locks 240 kg; independent backup release mechanisms 150 kg; local structural reinforcement and release fittings 160 kg; independent initiators/actuators 90 kg; release-state sensors and monitoring 65 kg; independent power/signalling and protected wiring 70 kg; fuel/energy isolation interfaces 55 kg; retention, inhibit and safety-interlock hardware 50 kg;

configuration growth allowance 20 kg. Avionics / Electrical / Hydraulic / Thermal / Control - 1,600 kg: Flight-control computers/control electronics 180 kg; flight-control actuators/servo hardware 210 kg; navigation/air-data/flight sensors 100 kg; communications/mission avionics 90 kg; electrical generation/distribution 190 kg; batteries/emergency electrical interfaces 90 kg; wiring/harnesses/connectors 170 kg; hydraulic pumps/reservoirs/lines/actuators 180 kg; thermal management/cooling 100 kg; environmental control/ventilation 70 kg; fire detection/warning/suppression interfaces 55 kg; health monitoring/data recording 45 kg; lighting/service/misc. 20 kg. Operating Fluids & Consumables - 320 kg: Engine/transmission oil and other operating fluids 180 kg; hydraulic fluid 35 kg; coolant/thermal working fluid 30 kg; other trapped/operating consumables 55 kg.

23. CRITICAL OPEN ITEMS - REV13

1.

Close the 26,000 kg mass model using component-level estimates and uncertainty.

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2.

Close the 8 occupant capsule geometry and bottom-up mass (Document 113).

3.

Select/size main rotor blade geometry, RPM, solidity, control system and transmission.

4.

Close tail rotor sizing against torque and directional-control requirements.

5.

Select candidate engines and close total installed power and OEI performance.

6.

Develop 1,000 km mission fuel model including reserves.

7.

Develop full 6-DOF aircraft/capsule/released-component model, prioritizing Stage S3 (§13) as the least-precedented item.

8.

Define the exact set of jettisonable non-rotor components for each emergency state.

9.

Prove safe separation trajectories and post-jettison control for each candidate component.

10.

Develop independent automatic and manual command architecture and common-cause analysis.

11.

Define Safety Kernel requirements and deterministic implementation boundary.

12.

Size the 3-primary/2-reserve canopy system and establish canopy failure tolerance (Document 159), including the low-altitude deployment envelope flagged as [OPEN - CRITICAL] in §14.

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13.

Evaluate optional residual-airframe parachute versus added mass, drag and deployment risk.

14.

Size the eight landing supports and energy absorbers, including asymmetric failures.

15.

Perform capsule structural, crash, fatigue and damage-tolerance analyses.

16.

Define occupant injury criteria and test instrumentation.

17.

Define emergency power and post-separation electrical architecture.

18.

Define fire, fuel isolation and thermal protection architecture.

19.

Define emergency egress, rescue, locator and communication provisions.

20.

Establish certification basis and verification matrix.

21.

Commission a professional prior-art search specifically against IP-A, IP-B and IP-C (§17, §19).

22.

Conduct independent structural sanity-check of the capsule mass budget (§8.1).

23.

Initiate the rotor-separation feasibility research track (Document 080) as a standalone, non-core workstream, should main-rotor separation be pursued in a future revision. These open items are not weaknesses to be hidden; they are the correct engineering gates that separate a credible conceptual IP baseline from an unsupported performance claim.

APPENDIX A - LICENSING PACKAGE (retained from Rev12,
unchanged in

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substance, cross-references updated) A.1 Executive Licensing Proposition
HSH Generation 1 is positioned as a licensable conceptual and architectural IP package centred on an occupied survival capsule that can become the final landing boundary in a catastrophic emergency. The Rev13 baseline defines a 26-ton- class single-main-rotor helicopter, a 26.0 m main rotor target, a 4.0 m tail-rotor target, a 1,000 km range target,

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controlled non-rotor selective separation, deterministic safety authorization, and multi-mode recovery. Core IP proposition: the occupied survival boundary is treated as a dedicated survival system rather than only as a conventional cabin. System-level differentiator: survival is organised as a chain of detection, authorization, selective separation, capsule protection, recovery and post-event survival. Commercial proposition: a licensee receives access to an architectural starting point and controlled IP-development framework, not a certified production helicopter. License value should be assessed against the reusable architecture, defined interfaces, the three primary candidate IP claims (§17.1), the disclosed risk register (§18) and verification roadmap, and technology-transfer potential. Status: [PROPOSED] Additive commercial/IP interpretation. No new performance claim is created by this section. A.2 Architectural Differentiation Map Occupied survival capsule as final landing boundary: HSH defines this as a core architectural requirement. State-dependent selective non-rotor separation: HSH explicitly defines release as an authorized emergency mode rather than an indiscriminate jettison command. Deterministic Safety Kernel: the baseline establishes a safety authorization boundary between complex prediction/decision functions and physical survival actuation. Six-layer survival chain: initiation,

independent backup/manual authority, selective separation, direct-impact survival and recovery/post-event survival are treated as distinct functional layers. Partial/asymmetric separation: the baseline explicitly treats incomplete release as a first-class emergency state requiring sensing, analysis and degraded recovery logic. Independent post- separation capability: the capsule domain includes independent emergency power, sensing, locator, communications and essential life-support provisions. Multi-mode recovery: controlled flight recovery and autorotation remain available where appropriate, while separation and capsule recovery form additional emergency modes. Status: [PROPOSED] Additive commercial/IP interpretation. A.3 IP Architecture / Value-Surface Map See §17 for the Rev13-narrowed claim set (IP-A, IP-B, IP-C) and §19 for the corresponding prior-art posture. Legal note: novelty, patentability, claim scope, validity and freedom-to-operate are not established by the baseline or this package. Professional prior-art and claim analysis remain required before filing. A.4 Licensee Due-Diligence & Maturity Roadmap A serious licensee should be able to distinguish what is defined from what is still open. Architecture: defined at conceptual baseline level, with the Rev13

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correction of the main-rotor dependency. Requirements: top-level requirements and targets are defined, with allocation and closure still required. Mass: 26,000 kg aircraft accounting and 7,000 kg capsule allocation are planning values requiring bottom-up closure (§8.1, §23). Dynamics: 6-DOF emergency dynamics, separation trajectories, partial/asymmetric release and parachute inflation remain open verification work, with Stage S3 (§13) as the priority gap. Separation: selective and staged architecture is defined conceptually and now explicitly excludes the main rotor; release mechanism reliability and safe trajectories require analysis and testing. Recovery: the 3+2 canopy architecture (§14) is a preliminary study concept requiring sizing, deployment and failure-case testing, with the low-altitude envelope as a flagged critical open item. Structures: capsule, attachments, landing supports and occupant protection require analytical and physical substantiation. Certification: the applicable certification basis and formal qualification evidence are not yet established. IP: three candidate surfaces are identified and prior-art-aware (§17, §19); novelty and freedom-to-operate require professional legal analysis. Gate logic: G0 through G8 as defined in §16, with G0 the only gate presently closed. Status: [PROPOSED] Additive commercial/IP interpretation. A.5 Staged Separation Architecture (commercial summary) For commercial readability, the corrected separation architecture (§13, §12) can be

expressed as three functional stages: Stage A - Hazard Reduction (isolate and separate designated non-rotor high-mass, rotating or fire-hazard components while maintaining controlled capsule load paths); Stage B - Capsule Liberation (release remaining designated airframe-to-capsule attachments so the occupied capsule can become the independent final landing boundary); Stage C - Clearance Verification & Recovery (positively determine capsule-clear status, identify residual attached components, and authorize the recovery system only within its validated envelope). The per-stage timing given in §13 remains a TARGET/OPEN development objective, not demonstrated performance. Partial and asymmetric release must remain explicitly represented in the state model at every stage.

Status: [PROPOSED] Additive commercial/IP interpretation. A.6 Technology Platform & Future Product Family The underlying survival architecture can be treated as a potential technology platform, while keeping future variants clearly separated from the Generation 1 engineering baseline: HSH-G1 Heavy (the present baseline); HSH Utility; HSH Rescue; HSH Expedition; HSH Special Mission. These future families are conceptual opportunities only, not implied to be engineered, tested, certified or commercially available. The reusable platform opportunity lies in

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architecture, interfaces, safety logic, survival-state management and verification methods, while vehicle-specific hardware must be re-engineered for each derivative. Status: [PROPOSED] Additive commercial/IP interpretation.

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A.7 Strategic Licensing Path Phase 1 - NDA: establish

confidentiality and

permitted evaluation scope. Phase 2 - Controlled Technical Data Room: provide the Rev13 baseline, this licensing package and the controlled document-tree material appropriate to the evaluation stage. Phase 3 - Technical/IP Due Diligence: review requirements, mass model, interfaces, separation concept, safety architecture, the Rev13 Risk Register (§18), verification roadmap and the three primary candidate IP surfaces (§17.1). Phase 4 - Exclusive Evaluation Option: if commercially appropriate, define a time-limited evaluation or option period with controlled technical access. Phase 5 - License Term Sheet: define territory, field of use, exclusivity,

milestones, economics, sublicensing and improvement rights. Phase 6 - Engineering Development Agreement: allocate responsibility for simulation, testing, qualification, certification and product definition. Phase 7 - Commercial License/Product Development: transition the controlled IP package into the licensee's engineering and industrialisation system. Commercial terms remain contractual matters and are not established by this engineering document. Status: [PROPOSED] Additive commercial/IP interpretation. A.8 Licensee's First 10 Questions

1.

What exactly is the IP? Three primary candidate surfaces (§17.1), explicitly narrowed from Rev12's nine, with supporting system architecture.

2.

What is actually defined today? The architecture, major interfaces, safety philosophy, emergency state machine, corrected non-rotor separation logic and verification structure, defined at conceptual-baseline level.

3.

What is not yet proven? Mass closure, detailed dynamics (especially Stage S3), separation reliability, parachute performance at low altitude, structural qualification, flight behaviour, certification and patentability - all disclosed explicitly in §18 and §23.

4.

What is the core differentiator? The occupied survival capsule combined with controlled, non-rotor, state- dependent separation and deterministic safety authorization.

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5.

Can the architecture scale? The survival principles can be considered a platform opportunity, but every derivative requires independent engineering closure and certification.

6.

What prevents uncontrolled separation? Mechanical retention, electrical/logic inhibits, independent command paths, defined arming, release-state sensing and safety preconditions (§6, §7.3).

7.

What happens if separation is incomplete? Partial/asymmetric release is explicitly treated as an emergency state at every stage, requiring state sensing and a defined degraded recovery strategy.

8.

How does AI interact with safety? AI may support detection, prediction and candidate-action ranking, while deterministic safety logic (§6) remains the sole authorization boundary - this is IP-A, the strongest claim in the package.

9.

How does the buyer know when the concept becomes credible? Evidence gates G0-G8 (§16) define the progression; Rev13 states plainly that only G0 is presently closed.

10.

What happens after licensing? The proposed path (§A.7) moves from controlled data-room evaluation into a term sheet and then a structured engineering-development programme. Status: [PROPOSED] Additive commercial/IP interpretation. END OF DOCUMENT 00 - HSH GENERATION 1 | REVISION 13 Capsule: 7,000 kg | Vehicle: 26 t | Main rotor: 26.0 m | Tail rotor: 4.0 m Founder: Amirbahador Atai | Not a certification approval

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PART II - REV14 ENGINEERING

EXPANSION

Purpose: increase engineering depth and reviewability while keeping Rev13's corrected safety posture.

The following sections are an additive engineering-development framework. They use the Rev13 baseline as the controlling conceptual architecture and the Rev12 001-500 document tree as the proposed evidence map. Items remain proposed until the referenced analysis, simulation, ground test, integrated test or certification activity closes them.

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1. REV14 CONFIGURATION AND DESIGN

PRINCIPLES

Rev14 keeps the Rev13 core principle that the survival case shall not depend on main-rotor separation. Main-rotor separation remains a deferred research track. The licensable core therefore concentrates on the occupied survival boundary, deterministic safety authorization, selective non-rotor separation, recovery, impact attenuation and post-event survival.

Configuration control is treated as a first-class engineering function. Changes to MTOW, rotor diameter, capsule mass, occupancy, release sequence, recovery architecture or safety-layer independence require a

formal impact assessment across mass properties, CG, inertia, dynamics, structures, controls, recovery and verification.

Configuration item

Rev14 control rule

Closure evidence

MTOW / design mass

Freeze only after bottom-up mass

model and reserve policy are

reconciled

Mass report + uncertainty

analysis

Rotor geometry

Maintain 26.0 m target until

performance study closes

Rotor sizing + power model

Capsule mass

7,000 kg planning target; no

closure claim

Bottom-up mass budget

Recovery

3 primary + 2 reserve study

architecture

Sizing + inflation dynamics +

failure testing

Separation

Non-rotor candidate set only in

core baseline

6-DOF + clearance + ground

demonstration

Safety authorization

Deterministic kernel remains

authoritative

Requirements +

software/hardware verification

2. SYSTEM REQUIREMENTS ARCHITECTURE

The Rev12 tree identifies a dedicated System Requirements Specification, system architecture, configuration management, interface control, risk management, safety management, V&V and certification-basis documents.

Rev14 turns those document families into a coherent requirements hierarchy.

Each top-level requirement should have a unique identifier, rationale, allocation, verification method, verification level, configuration applicability and evidence status. Requirements should not be written as

implicit descriptions buried inside narrative paragraphs.

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Requirement class

Example Rev14 allocation

Verification method

Mission

1,000 km range target, mission

profile dependent

Analysis + flight test

Mass

26,000 kg target with controlled

reserve

Analysis + weighing

Safety

Unsafe actuation inhibited unless

preconditions are satisfied

Analysis + test

Capsule

Independent emergency

functions after separation

Inspection + test

Separation

Release only within validated

envelope

Simulation + ground test

Recovery

Deployment restricted to

validated envelope

Analysis + test

Human authority

Defined manual command/inhibit

path

Human factors + test

3. MASS, CG AND INERTIA CLOSURE

Rev13 retains the 26,000 kg planning mass and 7,000 kg capsule allocation. Rev14 treats these as configuration-controlled hypotheses requiring bottom-up closure. The mass model should be maintained at component level, with uncertainty and growth tracked separately from nominal mass.

The same dataset should feed center-of-gravity and inertia calculations for normal flight, emergency configurations, capsule separation and

partial/asymmetric separation. A separation command cannot be evaluated independently of the resulting mass properties.

Mass case

Required outputs

Primary dependency

Normal aircraft

MTOW, empty weight, CG, inertia

Component mass model

Capsule separated

Capsule mass, CG, inertia

Capsule geometry and
equipment

Partial release

Residual mass and CG envelope

Release configuration matrix

Growth case

Margin and reserve consumption

Configuration control

Fuel extremes

CG movement and inertia

sensitivity

Fuel system model

4. MAIN ROTOR AND FLIGHT PERFORMANCE

DEVELOPMENT

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The 26.0 m main rotor produces a calculated disk area of 530.93 m² and Rev13's stated disk loading of approximately 48.97 kg/m². The document correctly labels hover-power calculations as study values. Rev14 expands the closure path rather than treating those values as final performance. Performance closure should include induced power, profile power, parasite power, transmission losses, engine power available, density altitude, temperature, rotor RPM, transient margins, OEI operation and autorotation behavior. The rotor disk loading trade should remain open until the dedicated performance study closes.

Analysis block

Key variables

Evidence

Hover

Mass, density, disk area, induced
velocity, FM

Power-required model

Climb

Power available - power required,
altitude, temperature

Performance analysis

Cruise

Drag, rotor efficiency, engine

SFC, fuel

Mission model

OEI

Remaining power, controllability,
density altitude

OEI analysis

Autorotation

Rotor inertia, descent state, flare
margin

Autorotation analysis

Vibration

Rotor dynamics, mast loads,
excitation

Dynamic analysis + test

5. PROPULSION, TRANSMISSION AND ENERGY

The baseline specifies two turboshaft-class engines as a proposed architecture. Rev14 requires explicit separation between engine selection, installation, transmission sizing, fuel system, thermal limits and emergency isolation. The propulsion system must be evaluated both for normal operation and for the survival sequence.

Fuel isolation is an enabling safety function. Isolation, venting, fire detection, engine shutdown, release sequencing and thermal management must be coordinated without assuming that any single subsystem will always respond correctly.

Function

Normal requirement

Emergency requirement

Engine control

Stable power and transient
response

Defined shutdown/isolation states

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Function

Normal requirement

Emergency requirement
Transmission
Torque, thermal and lubrication
limits
Safe degraded behavior
Fuel
Usable fuel and reserve
accounting
Isolation before/with release
where required
Thermal
Continuous heat rejection
Post-event thermal containment
OEI
Maintain defined safe envelope
No assumption of full
performance

6. SURVIVAL CAPSULE STRUCTURAL

ARCHITECTURE

Rev13 defines the capsule as the final occupied landing boundary, with a continuous CFRP roll-cage concept, metallic/titanium nodes, energy-absorbing seats, eight fixed supports and independent emergency functions. Rev14 expands the structural evidence chain.

The capsule program should establish load paths from ground contact through energy absorbers, lower structure, primary cell, floor, seat attachments and restraints. Structural cases should include vertical, longitudinal, lateral and combined loading, asymmetric touchdown, failed supports, slope and uneven terrain.

Subsystem

Primary question

Evidence required

Roll-cage

Can the cell maintain a survivable occupied volume?

FEA + structural test

Joints/nodes

Can interfaces transfer combined crash loads?

Coupon/component test

Floor

Can loads reach the intended absorbers without local failure?

FEA + test

Seats/restraints

Can occupant loads be managed
within criteria?

Dynamic test

Attachment fittings

Can normal and emergency load
paths be controlled?

Static/fatigue/test

7. EIGHT FIXED LANDING SUPPORTS

The eight supports are part of the survival system rather than conventional retractable landing gear. Rev14 treats their redundancy as an engineering hypothesis requiring asymmetric failure analysis, terrain interaction analysis and physical drop testing.

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The energy absorber should have a defined force-displacement response, inspection criteria and replacement philosophy. The analysis set should include all combinations that can materially change capsule attitude, peak loads or tipping resistance.

Case

Analysis focus

Required evidence

All supports available

Load sharing and stability

FEA + drop test

One support degraded

Redistribution and local overload

FEA + test

Multiple support degradation

Survivability boundary

Sensitivity study

Slope

Roll/tip tendency and load
transfer

Terrain simulation

Soft ground

Penetration and energy
dissipation

Ground interaction test

8. RECOVERY ARCHITECTURE: 3 PRIMARY +

2 RESERVE

Rev13 replaces the earlier undifferentiated eight-canopy concept with three primary and two reserve canopies. Rev14 defines the architecture as a study requiring sizing, inflation dynamics, riser loads, deployment sequencing, failure tolerance and low-altitude envelope definition.

No canopy count should be treated as proof of recovery performance. The system must be analyzed as a coupled capsule, harness, riser, canopy and atmosphere problem, including deployment transients and asymmetric failures.

Recovery state

Required assessment

Closure gate

Nominal primary deployment

Inflation, loads, descent state

Analysis + test

One primary unavailable

Remaining system authority

Simulation + test

Reserve deployment

Trigger logic and envelope

Analysis + test

Asymmetric inflation

Attitude/rate response

6-DOF + test

Low altitude

Available deployment time

Envelope definition

9. CONTROLLED RAPID SEPARATION

The separation architecture is selective and staged. Candidate non-rotor components include engines, selected transmission/drive components, tail rotor/drive components and isolated fuel hazards. Rev14 preserves the explicit exclusion of main-rotor separation from the core safety case.

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Each release event requires positive retention in normal flight, electrical/logic inhibition, independent command paths, release-state sensing, energy isolation, asymmetric-release detection, clearance verification and a defined fallback.

Stage

Function

Evidence

S0

Validate state, arm authorized

path, isolate hazards

Logic verification

S1

Fuel/fire-hazard actions

Subsystem test

S2

Non-rotor component separation

Dynamic + ground test

S3

Capsule release and clearance

6-DOF + integrated test

S4

Recovery authorization

Envelope verification

10. DETERMINISTIC SAFETY KERNEL

The Safety Kernel is the authorization boundary between complex detection/decision logic and physical survival actuation. Rev14 strengthens the requirement that AI, predictive models and optimizers may assist monitoring but cannot bypass deterministic safety preconditions.

Kernel inputs should include altitude, airspeed, attitude and rates, rotor state, engine/fire/fuel state, structural health, capsule attachment health, separation clearance, recovery-system availability, sensor validity, power availability and human authority state.

Kernel decision

Minimum design question

Verification

Authorize release

Are all mandatory preconditions true?

Formal/structural logic review + test

Inhibit release

Can an unsafe state block actuation?

Fault injection

Fallback

What happens if a sensor/path is lost?

Failure testing

Human override

Is authority bounded and deterministic?

HMI + system test

Audit

Can the event be reconstructed?

Data-recording test

11. EMERGENCY STATE MACHINE AND

MODE LOGIC

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Rev13 defines explicit states from Normal Flight through Anomaly Detection, Confirmation, Impact Assessment, Recovery Attempt, Survival Authorization, Staged Separation, Capsule Clearance, Recovery Deployment, Descent Management, Landing Attenuation, Post-Landing Survival and Audit.

Rev14 expands this into a state-transition verification discipline. Every transition should identify entry conditions, exit conditions, inhibited transitions, sensor validity requirements, crew authority, time constraints and degraded behavior.

State

Entry condition

Exit / next state

Normal

Nominal monitoring

Anomaly detected or normal continuation

Confirmation

Potential abnormality

Confirmed or rejected condition

Recovery attempt

Safe conventional recovery available

Recovery success or survival authorization

Survival authorization

Defined catastrophic conditions

Staged separation or inhibited state

Capsule clearance

Release executed

Recovery deployment authorization

Post-landing

Ground contact / stabilized

Rescue and audit complete

12. FAILURE MODES, FMEA AND

COMMON-CAUSE CONTROL

Rev13 identifies unintended separation, partial separation, asymmetric separation, release failure, sensor loss, automatic detection failure, power loss, communication loss, AI/model failure, fire during separation, released-component collision, parachute deployment outside envelope and landing-support failure.

Rev14 establishes these as seed cases for FMEA/FMECA, fault-tree analysis, common-cause analysis, zonal safety analysis and particular-risk analysis. Independence must be demonstrated against common power, wiring, software, sensors, structure, fire zones and environmental exposure.

Failure mode

Immediate effect

Design response

Unintended release

Loss of structural attachment

Mechanical locks + inhibits +
monitoring

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Failure mode

Immediate effect

Design response

Partial release

Asymmetric mass/inertia

State sensing + degraded
recovery

Sensor disagreement

Unsafe authorization risk

Validity logic + diverse sensing

Power loss

Loss of survival functions

Independent emergency power

AI/model failure

Bad recommendation

Kernel remains authoritative

Fire

Loss of structure or fuel hazard

Isolation + containment +
sequencing

13. 6-DOF DYNAMICS AND SEPARATION

MODEL

The Rev12 tree explicitly calls for pre-release aircraft dynamics, post-separation capsule dynamics, released-component trajectory models, partial/asymmetric release cases, parachute inflation transients and ground-contact kinematics. Rev14 consolidates these into one model hierarchy.

The model should represent mass properties, aerodynamic forces, rotor effects, control authority, release impulses, component trajectories, capsule attitude/rates and recovery-system deployment. Uncertainty should be propagated rather than hidden.

Model

Minimum outputs

Validation path

Aircraft pre-release

Position, attitude, rates, control margin

Bench/simulation correlation

Capsule release

6-DOF trajectory and clearance

Dynamic simulation + test

Released components

Collision envelope

Trajectory test/data

Partial release

CG/inertia and stability changes

Fault-injection simulation

Parachute inflation

Transient loads and attitude

Inflation testing

Ground contact

Support loads and tipping

Drop/terrain tests

14. VERIFICATION, VALIDATION AND

EVIDENCE GATES

Rev13 defines G0-G8 progression. Rev14 makes the gates evidence-based rather than document-based. A gate should close only when requirements, models, test data, discrepancies and configuration records are consistent.

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The proposed sequence is G0 Architecture Freeze, G1 Analytical Closure,

G2 Subsystem Demonstration, G3 Integrated Ground Demonstration, G4 Dynamic/6-DOF Simulation Validation, G5 Structural/Crash Testing, G6 Recovery System Testing, G7 Flight Development and G8 Certification/Qualification.

Gate

Primary evidence

Exit condition

G0

Controlled architecture +
requirements

Baseline frozen

G1

Mass, CG, performance, safety
analyses

Analytical closure acceptable

G2

Component/subsystem tests

Critical functions demonstrated

G3

Integrated ground sequence

End-to-end ground evidence

G4

Validated dynamics models

Model correlation and uncertainty
bounded

G5

Structural/crash evidence

Defined structural cases closed

G6

Recovery evidence

Recovery envelope demonstrated

G7

Flight development evidence

Flight envelope progressively
established

G8

Compliance evidence

Certification/qualification basis
satisfied

15. MASTER VERIFICATION MATRIX

The Rev12 tree identifies a Master Verification Matrix, Requirements Traceability Matrix, Verification Evidence Register, nonconformance process and review entry/exit criteria. Rev14 makes these the central

evidence-control mechanism.
Requirement evidence field
Required content
Requirement ID
Unique controlled identifier
Source
Parent requirement / architecture reference
Allocation
System / subsystem owner
Verification method
Analysis, inspection, demonstration, test
Configuration
Applicable hardware/software revision
Result
Pass, fail, conditional, open
Evidence
Report, dataset, test record or approved analysis
Discrepancy
Open NCR/deviation if applicable
Approval
Responsible engineering authority

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16. CERTIFICATION AND COMPLIANCE

READINESS

Rev13 explicitly states that the document is not certification approval. Rev14 therefore separates conceptual architecture from certification evidence. The Rev12 tree includes a CS-29-oriented certification basis, means-of-compliance matrix, compliance checklist, flight-test instrumentation plan, continued-airworthiness plan and flight-manual development plan.

The correct objective at this stage is traceability: every future certification claim should have an identified rule or certification objective, means of compliance, verification method, responsible owner and evidence record.

Certification work product

Rev14 role

Certification Basis

Identify applicable requirements and assumptions

Means of Compliance Matrix

Map each requirement to evidence

Compliance Checklist

Track status and deviations

Type Design Definition

Transition from concept to controlled design

Continued Airworthiness

Define future inspection/maintenance evidence

17. HUMAN FACTORS AND CREW

AUTHORITY

Rev13 preserves human emergency authority. Rev14 expands this into a human-machine interface workstream. The crew should understand system state, inhibited actions, available recovery modes and the consequences of manual intervention.

Emergency controls should be physically and logically separated from routine controls where justified. Human authority must remain bounded by physical safety interlocks so that manual action cannot silently defeat mandatory safety conditions.

HMI element

Design objective

Verification

Emergency state display

Unambiguous state awareness

Simulator + usability test

Manual initiation

Fast, deliberate action

Human factors test

Inhibit control

Clear consequence and scope

Procedure validation

Fault indication

Distinguish sensor disagreement
from confirmed hazard

Scenario test

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HMI element

Design objective

Verification

Post-event status

Show capsule/recovery condition

Ground test

18. SOFTWARE, AI AND DIGITAL-TWIN

BOUNDARIES

The Rev12 tree contains AI decision-support constraints, health

monitoring, digital-twin estimation and predictive fault detection. Rev13 establishes the key boundary: predictive or AI functions may support detection and decision support, but deterministic safety logic remains authoritative.

Rev14 therefore separates advisory computation from safety authorization. Model confidence is not equivalent to authorization. Any AI-derived recommendation entering a safety workflow should be traceable, bounded and testable against defined failure cases.

Function

Permitted role

Safety boundary

Anomaly detection

Advisory / monitoring

Cannot authorize unsafe
actuation alone

Prediction

Risk estimation

Must feed bounded logic

Optimization

Mission/performance support

Cannot bypass limits

Digital twin

State estimation / analysis

Validation required before
operational reliance

Safety kernel

Authorization

Deterministic and independently
verified

19. IP VALUE SURFACES AND PRIOR-ART

POSTURE

Rev13 narrows the strongest candidate IP surfaces to a smaller set centered on the occupied survival boundary, state-dependent selective separation and deterministic safety authorization, with supporting architecture. The Rev12 tree also identifies IP elements 0001-0009, prior-art search, freedom-to-operate assessment, claim-drafting guidance and invention disclosure work.

No novelty, patentability or freedom-to-operate conclusion should be inferred from the engineering baseline. The correct Rev14 action is controlled invention disclosure followed by professional prior-art and FTO analysis.

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IP surface

Technical evidence needed

Legal work

Occupied survival boundary

Architecture + structural
evidence

Prior-art search + claim drafting

State-dependent separation

State machine + release
evidence

Prior-art/FTO

Deterministic safety kernel

Requirements + verification
evidence

Prior-art/FTO

Six-layer survival architecture

Independence analysis

Landscape study

Eight-support attenuation

Load path + test data

Novelty assessment

20. LICENSING AND TECHNICAL DUE

DILIGENCE PACKAGE

The licensing package should make the maturity state transparent. A prospective licensee should receive controlled access to the baseline, document tree, requirements, risk register, verification roadmap and relevant technical evidence appropriate to the evaluation stage.

The commercial path should separate technical diligence from contractual decisions. Engineering documents define technical scope and evidence status; legal and commercial agreements define territory, field of use, exclusivity, economics, milestones and improvement rights.

Diligence phase

Primary material

Decision output

NDA / controlled access

Document control + scope

Authorized evaluation

Technical review

Architecture + requirements

Issue list

Safety review

Risk/FMEA/verification

Safety evidence gaps
IP review
Invention disclosures + prior art
status
IP work plan
Commercial review
Licensing package
Term-sheet negotiation

21. ENGINEERING RISK REGISTER - REV14

SEED

The risk register is not a declaration of safety. It is a mechanism for identifying where the concept can fail and where evidence must be generated before a safety claim is made.

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Risk

Current status

Closure activity

Priority

Mass closure

OPEN

Bottom-up mass model

+ weighing plan

Critical

Capsule structural

closure

OPEN

FEA +

component/static/crash

tests

Critical

Separation dynamics

OPEN

6-DOF + clearance

analysis

Critical

Recovery performance

OPEN

Canopy sizing + inflation

testing

Critical

Partial/asymmetric

release

OPEN

Fault cases + dynamic

simulation

Critical

Safety independence

OPEN

CCA + FMEA/FHA/PSSA

Critical

Certification basis

OPEN

Authority engagement +

compliance map

High

Patentability/FTO

NOT ESTABLISHED

Professional IP search

High

22. REV14 DESIGN REVIEW CHECKLIST

A mature review should challenge the architecture rather than merely confirm that documents exist. The following checklist is intended for G0/G1 design reviews.

Review question

Expected evidence

Does every top-level requirement have an owner?

Requirements allocation

Does every mass number have a source/status?

Mass database

Does every safety function have a verification path?

V&V matrix

Can partial separation be detected?

State sensing design

Can a common power failure defeat multiple safety layers?

CCA

Can the capsule survive asymmetric touchdown?

Structural analysis/test plan

Is recovery deployment envelope defined?

Parachute analysis

Is AI prevented from bypassing authorization?

Kernel interface specification

Can an emergency event be reconstructed?

Immutable event record design

23. REV14 MATURITY ROADMAP

Rev14 is best treated as a bridge between a strong conceptual/IP baseline and an engineering program capable of producing objective evidence. The

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roadmap below follows the document families already identified in the Rev12 Master Engineering Document Tree.

Period / phase

Primary output

Gate

Architecture control

Requirements, interfaces,
configuration baseline

G0

Analytical closure

Mass, CG, inertia, power,
dynamics, safety analyses

G1

Subsystem design

Capsule, supports, separation,
recovery, avionics

G2 preparation

Ground demonstration

Integrated release/recovery
sequence

G3

Model validation

6-DOF correlation and
uncertainty

G4

Structural/recovery tests

Crash, drop, canopy and
component evidence

G5/G6

Flight development

Progressive envelope expansion

G7

Qualification/certification

Compliance evidence and
controlled type design

G8

24. REV14 ENGINEERING INTEGRITY

STATEMENT

This document remains a conceptual engineering and IP development baseline. TARGET, ASSUMPTION, STUDY, CALC and OPEN values do not become demonstrated performance merely because they appear in a detailed document. The enlarged technical framework is intended to make future verification more disciplined, not to substitute documentation for evidence.

The strongest version of HSH Generation 1 is therefore not the version with the most impressive claims. It is the version in which every important claim has a visible path from requirement to model to test to controlled evidence.

Status class

Meaning

LOCKED CONCEPT

Architecture intentionally retained; implementation still requires verification

REQ

Proposed requirement for allocation and verification

TARGET

Development objective, not demonstrated

ASSUMPTION / STUDY

Working preliminary engineering assumption

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Status class

Meaning

CALC

Calculated from stated assumptions; independently reviewable

OPEN

Requires analysis, simulation, test or design decision

OPEN - CRITICAL

Open item materially affecting the core safety case

DEFERRED - RESEARCH TRACK

Explicitly outside the licensable core pending independent research

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APPENDIX B - REV14

CONTROLLED DOCUMENT-TREE

ALIGNMENT

The Rev12 Master Engineering Document Tree contains 500 sequential documents covering requirements, safety, mass properties, structures, rotor systems, propulsion, capsule, separation, avionics, performance, simulation, testing, verification, certification, IP and program management. Rev14 uses that tree as the proposed maturity map rather than reproducing all 500 entries inside Document 00.

Document family 001-022

Document family 023-044

Document family 111-154

Document family 157-220

Document family 221-264

Document family 265-330

Document family 331-374

Document family 399-430

Document family 461-500

END OF HSH-00 REVISION 14

Founder: Amirbahador Atai

Conceptual / IP Baseline | Proprietary Concept Development

Material

PART II - REV15 FINAL ENGINEERING CONTROL

ADDENDUM

5. Master configuration record

Configuration item

Rev15 baseline

Status

Aircraft

HSH Heavy Safe Helicopter

LOCKED

Founder

Amirbahador Atai

CONTROLLED

Generation

Generation 1

LOCKED

Document

HSH-00

LOCKED

Revision

Rev15

BASELINE

Capsule occupancy

8 persons maximum

LOCKED LIMIT

Vehicle mass

26,000 kg

TARGET / OPEN

Capsule allocation

7,000 kg

PLANNING / OPEN

Main rotor

26.0 m

TARGET / OPEN

Tail rotor

4.0 m

TARGET / OPEN

Range

1,000 km

TARGET / OPEN

Landing supports

8 fixed energy-absorbing supports

CONCEPT / OPEN

Recovery

3 primary + 2 reserve canopies

STUDY / OPEN

Core separation

Selective, staged, non-rotor

CONCEPT / OPEN

Main-rotor separation

Excluded from core safety case

DEFERRED RESEARCH

Safety authorization

Deterministic Safety Kernel

CONCEPT / OPEN

6. Closure priority matrix

Priority

Work package

Required evidence

P0

Maximum-8-person capsule closure

Bottom-up mass, geometry, CG/inertia, structural

and occupant-protection evidence

P0

Stage S3 capsule release and clearance

6-DOF dynamics, clearance analysis, mechanism tests and integrated ground evidence

P0

Recovery envelope

Canopy sizing, inflation dynamics, loads, low-altitude boundary and failure-case testing

P0

Capsule crashworthiness

FEA, component tests, drop/crash testing and occupant-protection evidence

P0

Safety independence

FHA/PSSA/FMEA/CCA and fault-injection evidence

P1

26,000 kg mass closure

Component database, weighing plan, uncertainty and growth control

P1

Rotor and propulsion closure

Rotor sizing, power, OEI, transmission and thermal analyses

P1

Requirements traceability

Unique IDs, allocation, verification method and evidence register

P1

Certification basis

Applicable requirements, means of compliance and formal engagement

P1

IP due diligence

Professional prior-art, claim-scope and FTO analysis

7. Final Generation 1 baseline statement

HSN-00 Rev15 establishes the final Generation 1 conceptual master baseline for the HSH Heavy Safe Helicopter. The survival architecture centers on an occupied survival capsule, deterministic safety authorization, selective non-rotor separation, multi-mode recovery, fixed energy-absorbing landing supports and post-event survival capability.

The capsule occupancy limit is fixed at 8 persons maximum for Generation 1. Future technical work shall not silently alter this value. Any proposed change requires formal configuration control and impact analysis.

The document deliberately distinguishes architecture from evidence. Mass closure, 6-DOF dynamics, separation trajectories, recovery performance, crashworthiness, safety independence, certification and patentability remain evidence-generating work packages. The appropriate next step is therefore not another expansion of Document 00. It is controlled execution of the HSH Generation 1 engineering document tree, with every major claim linked to a requirement, analysis, simulation, test result and configuration-controlled evidence record.

END OF HSH-00 REVISION 15

HSH Heavy Safe Helicopter | Generation 1 | Founder: Amirbahador Atai

REV16 DOCUMENT-CONTROL CLOSURE

Rev16 is a formatting and document-control revision of HSH-00 Rev15. Its principal purpose is to remove unnecessary blank pages and improve continuous page utilization for technical review and due-diligence handling. The Rev15 substantive baseline is retained.

Controlled identity: HSH Heavy Safe Helicopter | HSH-00 | Generation 1 | Founder: Amirbahador Atai.

Occupancy control: maximum 8 persons in the Generation 1 survival capsule. No 10-person or 12-person configuration is part of this baseline.

Baseline status: Final Generation 1 Conceptual / IP Master Engineering Baseline. Future technical closure belongs in the controlled HSH Generation 1 document tree.

END OF HSH-00 REVISION 16

HSH Heavy Safe Helicopter | Generation 1 | Founder: Amirbahador Atai